

Packed Qwen-Pair Training Row

View 1 (original)

View 2 (rewrite)

Original 1

Rewrite 1

Original 2

Rewrite 2

Typical row: 3-4 paired segments, ~150 words

Three Masking/Credit Strategies on the Same Row

Ordinary WWM (15%)

Random 15% of all positions masked and predicted
(applied to non-pair rows and as ordinary-mask preservation target)

*Dense masking removes within-view completion clues,
making cross-view correspondence the most economical
explanation for the sparse targets*

Dense mask + sparse label (M,S)

All detected content groups in View 2 are masked (132,283 groups)
Loss computed only on a deterministic sparse subset (21,479 groups)
View 1 (original) remains fully visible throughout

Dense mask + dense label (M,M)

Same dense masking of View 2 content groups
Loss computed on ALL masked positions (132,283 groups)